

NetworkAnalysis_PositivePsychology

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Descriptive statistics

```
#import dataset
library(readxl)
PPI_dataset <- read_excel("/Users/hyeinkim/Documents/N741/milestone/PPI_dataset.xlsx")

library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)

PPI_dataset %>%
  summarise(across(everything(), ~ mean(is.na(.) | . == -999) * 100)) %>%
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Missing_Percentage") %>%
  mutate(Missing_Percentage = round(Missing_Percentage, 2)) %>%
  arrange(desc(Missing_Percentage))
```

```
## # A tibble: 175 x 2
##   Variable      Missing_Percentage
##   <chr>          <dbl>
## 1 AHS_PostTotal      3
## 2 Flourish_PostTotal  2
## 3 BMPN_Sat_Rel_Post  2
## 4 BMPN_Sat_Com_Pre   2
## 5 Usage              2
## 6 AHS_POST_2          2
```

```
## 7 AHS_POST_4 2
## 8 Flourish_PreTotal 1
## 9 CESD_PostTotal 1
## 10 PANAS_Pos_Post 1
## # i 165 more rows
```

```
#missing data handling
# convert "-999" to NA in the dataset
PPI_dataset[PPI_dataset == -999] <- NA

# remove row with NA
PPI_dataset <- na.omit(PPI_dataset)
```

```
#name categorical variables
```

```
PPI_dataset$Income <- factor(PPI_dataset$Income)
PPI_dataset$Sex <- factor(PPI_dataset$Sex)

PPI_dataset$Income <- factor(PPI_dataset$Income,
                             levels = c(1, 2, 3, 4, 5),
                             labels = c("<$9999",
                                         "$10,000-$29,999",
                                         "$30,000-$59,999",
                                         "$60,000-$89,999",
                                         ">$90,000"))

PPI_dataset$Sex <- factor(PPI_dataset$Sex,
                          levels = c(1, 2),
                          labels = c("Male", "Female"))
```

```
#Basic Descriptive Statistics, Table 1
```

```
summary(PPI_dataset[, c("Flourish_PreTotal", "CESD_PreTotal", "AHS_PreTotal",
                        "BMPN_Sat_Aut_Pre", "BMPN_Sat_Rel_Pre", "BMPN_Sat_Com_Pre",
                        "PANAS_Pos_Pre", "AGE", "Income", "Sex")])
```

```
## Flourish_PreTotal CESD_PreTotal AHS_PreTotal BMPN_Sat_Aut_Pre
## Min. :19.00 Min. : 0.000 Min. :14.00 Min. : 8.00
## 1st Qu.:36.50 1st Qu.: 3.500 1st Qu.:19.50 1st Qu.:13.00
## Median :41.00 Median : 8.000 Median :22.00 Median :15.00
## Mean :39.88 Mean : 7.843 Mean :21.73 Mean :14.65
## 3rd Qu.:44.00 3rd Qu.:11.000 3rd Qu.:24.00 3rd Qu.:16.00
## Max. :54.00 Max. :25.000 Max. :30.00 Max. :20.00
## BMPN_Sat_Rel_Pre BMPN_Sat_Com_Pre PANAS_Pos_Pre AGE
## Min. : 7.0 Min. : 7.00 Min. :13.00 Min. :19.00
## 1st Qu.:15.0 1st Qu.:12.00 1st Qu.:22.00 1st Qu.:21.00
## Median :16.0 Median :15.00 Median :27.00 Median :22.00
## Mean :15.4 Mean :14.33 Mean :26.95 Mean :22.86
## 3rd Qu.:16.0 3rd Qu.:16.00 3rd Qu.:32.00 3rd Qu.:24.00
## Max. :20.0 Max. :20.00 Max. :48.00 Max. :34.00
## Income Sex
## <$9999 : 8 Male :25
## $10,000-$29,999:30 Female:58
```

```
## $30,000-$59,999: 37
## $60,000-$89,999: 6
## >$90,000 : 2
##
```

```
#Mean and SD
library(table1)
```

```
##
## Attaching package: 'table1'
```

```
## The following objects are masked from 'package:base':
##
## units, units<-
```

```
table1(~Flourish_PreTotal+CESD_PreTotal+AHS_PreTotal+BMPN_Sat_Aut_Pre+BMPN_Sat_Rel_Pre+BMPN_Sat_Com_Pre
```

```
## Get nicer 'table1' LaTeX output by simply installing the 'kableExtra' package
```

	Overall
	(N=83)
Flourish_PreTotal	
Mean (SD)	39.9 (7.00)
Median [Min, Max]	41.0 [19.0, 54.0]
CESD_PreTotal	
Mean (SD)	7.84 (5.57)
Median [Min, Max]	8.00 [0, 25.0]
AHS_PreTotal	
Mean (SD)	21.7 (3.24)
Median [Min, Max]	22.0 [14.0, 30.0]
BMPN_Sat_Aut_Pre	
Mean (SD)	14.7 (2.28)
Median [Min, Max]	15.0 [8.00, 20.0]
BMPN_Sat_Rel_Pre	
Mean (SD)	15.4 (2.48)
Median [Min, Max]	16.0 [7.00, 20.0]
BMPN_Sat_Com_Pre	
Mean (SD)	14.3 (2.52)
Median [Min, Max]	15.0 [7.00, 20.0]
PANAS_Pos_Pre	
Mean (SD)	27.0 (7.21)
Median [Min, Max]	27.0 [13.0, 48.0]
AGE	
Mean (SD)	22.9 (3.35)
Median [Min, Max]	22.0 [19.0, 34.0]
Income	
<\$9999	8 (9.6%)
\$10,000-\$29,999	30 (36.1%)
\$30,000-\$59,999	37 (44.6%)
\$60,000-\$89,999	6 (7.2%)

	Overall
>\$90,000	2 (2.4%)
Sex	
Male	25 (30.1%)
Female	58 (69.9%)

#Correlation among the psychological constructs

```
library(dplyr)
cor_data <- PPI_dataset %>%
  select(Flourish_PreTotal, CESD_PreTotal, AHS_PreTotal,
         BMPN_Sat_Aut_Pre, BMPN_Sat_Rel_Pre, BMPN_Sat_Com_Pre, PANAS_Pos_Pre)

cor_matrix <- cor(cor_data, use = "complete.obs", method = "pearson")
cor_matrix
```

```
##               Flourish_PreTotal CESD_PreTotal AHS_PreTotal BMPN_Sat_Aut_Pre
## Flourish_PreTotal      1.0000000    -0.5562166     0.7161300     0.6957561
## CESD_PreTotal          -0.5562166     1.0000000    -0.5203264    -0.4612272
## AHS_PreTotal           0.7161300    -0.5203264     1.0000000     0.7060230
## BMPN_Sat_Aut_Pre       0.6957561    -0.4612272     0.7060230     1.0000000
## BMPN_Sat_Rel_Pre       0.4369726    -0.2154051     0.3015373     0.3849476
## BMPN_Sat_Com_Pre       0.7268609    -0.6445542     0.7275077     0.7379862
## PANAS_Pos_Pre         0.5320593    -0.2989516     0.5360076     0.4502693
##               BMPN_Sat_Rel_Pre BMPN_Sat_Com_Pre PANAS_Pos_Pre
## Flourish_PreTotal      0.4369726     0.7268609     0.5320593
## CESD_PreTotal          -0.2154051    -0.6445542    -0.2989516
## AHS_PreTotal           0.3015373     0.7275077     0.5360076
## BMPN_Sat_Aut_Pre       0.3849476     0.7379862     0.4502693
## BMPN_Sat_Rel_Pre       1.0000000     0.3709115     0.2173045
## BMPN_Sat_Com_Pre       0.3709115     1.0000000     0.5120808
## PANAS_Pos_Pre         0.2173045     0.5120808     1.0000000
```

```
library(corr)
cor_data %>%
  correlate() %>%
  stretch() %>%
  arrange(desc(r))
```

```
## Correlation computed with
## * Method: 'pearson'
## * Missing treated using: 'pairwise.complete.obs'
```

```
## # A tibble: 49 x 3
##       x                y                r
##   <chr>          <chr>          <dbl>
## 1 BMPN_Sat_Aut_Pre BMPN_Sat_Com_Pre 0.738
## 2 BMPN_Sat_Com_Pre BMPN_Sat_Aut_Pre 0.738
## 3 AHS_PreTotal     BMPN_Sat_Com_Pre 0.728
## 4 BMPN_Sat_Com_Pre AHS_PreTotal     0.728
## 5 Flourish_PreTotal BMPN_Sat_Com_Pre 0.727
```

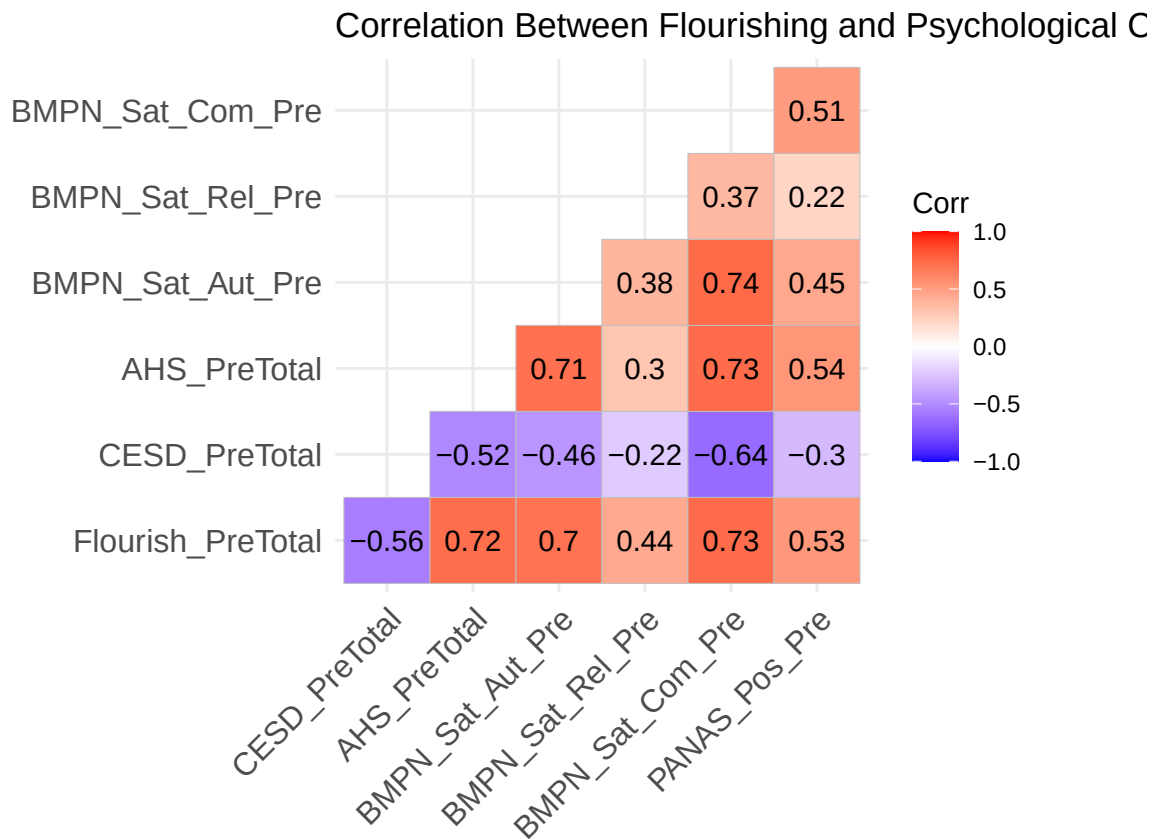
```
## 6 BMPN_Sat_Com_Pre Flourish_PreTotal 0.727
## 7 Flourish_PreTotal AHS_PreTotal 0.716
## 8 AHS_PreTotal Flourish_PreTotal 0.716
## 9 AHS_PreTotal BMPN_Sat_Aut_Pre 0.706
## 10 BMPN_Sat_Aut_Pre AHS_PreTotal 0.706
## # i 39 more rows
```

#Figure 1. Heatmap plot

```
library(ggcorrplot)
```

```
## Loading required package: ggplot2
```

```
ggcorrplot(cor_matrix,
  lab = TRUE,
  type = "lower",
  title = "Correlation Between Flourishing and Psychological Constructs")
```



Network analysis

Select only PANAS_pro and Flourishing

```
library(bootnet)
```

```
## This is bootnet 1.6
```

```
## For questions and issues, please see github.com/SachaEpskamp/bootnet.
```

```
library(qgraph)
library(dplyr)

#Preprocessing

#select PANAS_positive and Flourishing scale

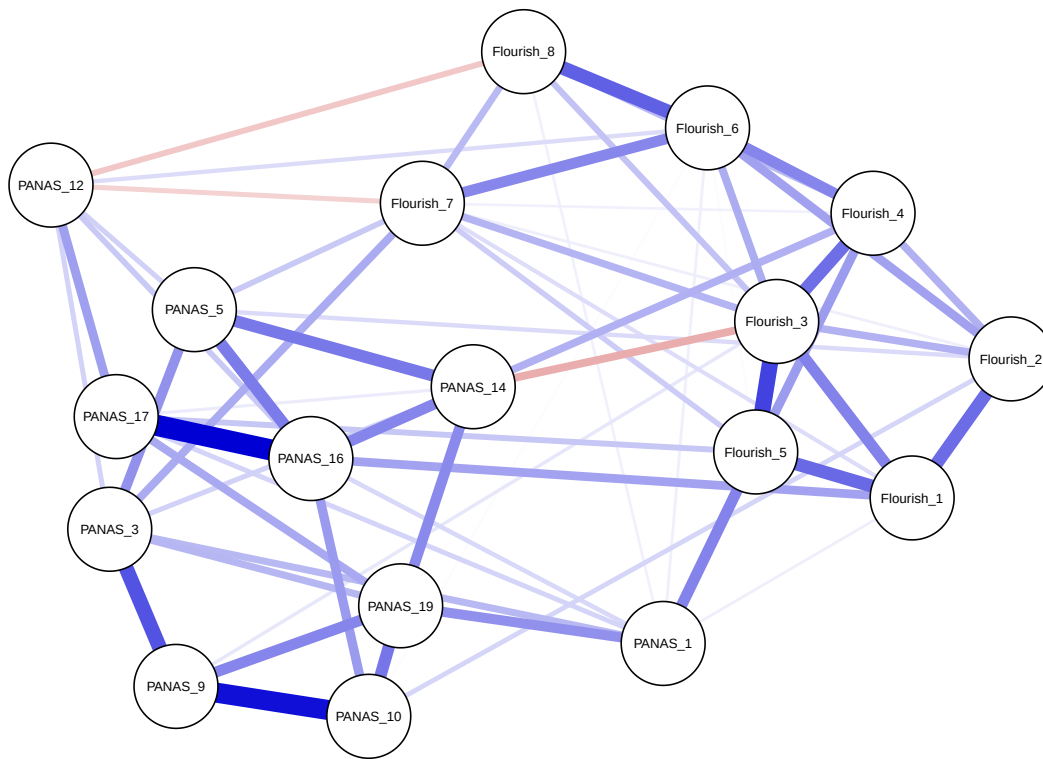
network_data1 <- PPI_dataset[, c("PANAS_1", "PANAS_3", "PANAS_5", "PANAS_9", "PANAS_10", "PANAS_12", "P
network_data1 <- na.omit(network_data1)
network_data1 <- network_data1[, sapply(network_data1, function(x) sd(x, na.rm = TRUE) != 0)]

library/bootnet)
network_model1 <- estimateNetwork(network_data1,
                                default = "EBICglasso",
                                corMethod = "cor_auto")

## Estimating Network. Using package::function:
##   - qgraph::EBICglasso for EBIC model selection
##   - using glasso::glasso
##   - qgraph::cor_auto for correlation computation
##   - using lavaan::lavCor

## Variables detected as ordinal: PANAS_1; PANAS_3; PANAS_5; PANAS_9; PANAS_10; PANAS_12; PANAS_14; PAN

plot(network_model1)
```



#flourishing and AHS (Adult Hope Scale)

```
network_data2 <- PPI_dataset[, c("AHS_1", "AHS_2", "AHS_3", "AHS_4", "AHS_5", "AHS_6", "AHS_7", "AHS_8")

network_data2 <- na.omit(network_data2)
network_data2 <- network_data2[, sapply(network_data2, function(x) sd(x, na.rm = TRUE) != 0)]

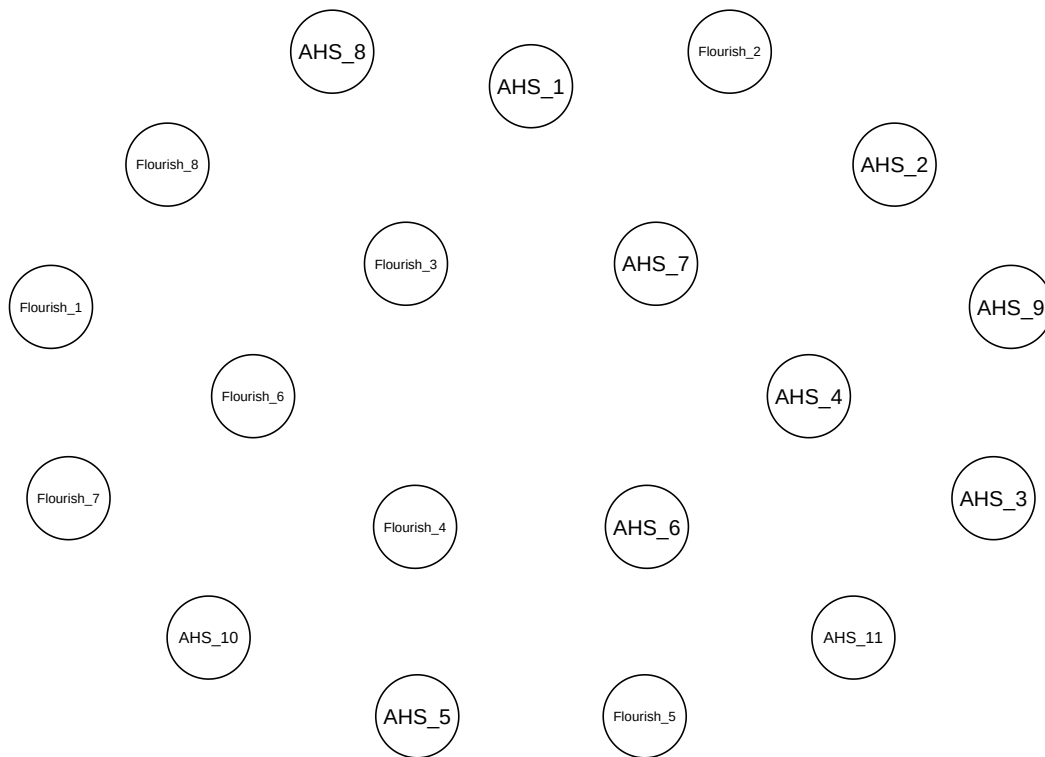
library(bootnet)
library(qgraph)
library(dplyr)

network_model2 <- estimateNetwork(network_data2,
                                  default = "EBICglasso",
                                  corMethod = "cor")

## Estimating Network. Using package::function:
## - qgraph::EBICglasso for EBIC model selection
## - using glasso::glasso

## An empty network was selected to be the best fitting network. Possibly set 'lambda.min.ratio' higher

plot(network_model2)
```



“An empty network was selected to be the best fitting network.” → What should I do to solve this problem?

#CESD recoding evaluation

```
# 1. select only necessary variables
cesd_vars <- PPI_dataset %>%
  select(CESD_PreTotal, CESD_5, CESD_8)
```

```
# 2.correlation test
cor.test(cesd_vars$CESD_5, cesd_vars$CESD_PreTotal) # CESD_5 vs Total
```

```
##
## Pearson's product-moment correlation
##
## data: cesd_vars$CESD_5 and cesd_vars$CESD_PreTotal
## t = 8.8139, df = 81, p-value = 1.854e-13
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.5700169 0.7953434
## sample estimates:
## cor
## 0.6996826
```

```
cor.test(cesd_vars$CESD_8, cesd_vars$CESD_PreTotal) # CESD_8 vs Total
```

```
##
```



```
## Pearson's product-moment correlation
##
## data:  cesd_vars$CESD_8 and cesd_vars$CESD_PreTotal
## t = 13.49, df = 81, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.750896 0.888186
## sample estimates:
##      cor
## 0.8318575
```

```
# make new recoded variables
cesd_vars <- cesd_vars %>%
  mutate(
    CESD_5_R = 3 - CESD_5,
    CESD_8_R = 3 - CESD_8
  )

# recheck the correlation
cor.test(cesd_vars$CESD_5_R, cesd_vars$CESD_PreTotal)
```

```
##
## Pearson's product-moment correlation
##
## data:  cesd_vars$CESD_5_R and cesd_vars$CESD_PreTotal
## t = -8.8139, df = 81, p-value = 1.854e-13
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.7953434 -0.5700169
## sample estimates:
##      cor
## -0.6996826
```

```
cor.test(cesd_vars$CESD_8_R, cesd_vars$CESD_PreTotal)
```

```
##
## Pearson's product-moment correlation
##
## data:  cesd_vars$CESD_8_R and cesd_vars$CESD_PreTotal
## t = -13.49, df = 81, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.888186 -0.750896
## sample estimates:
##      cor
## -0.8318575
```

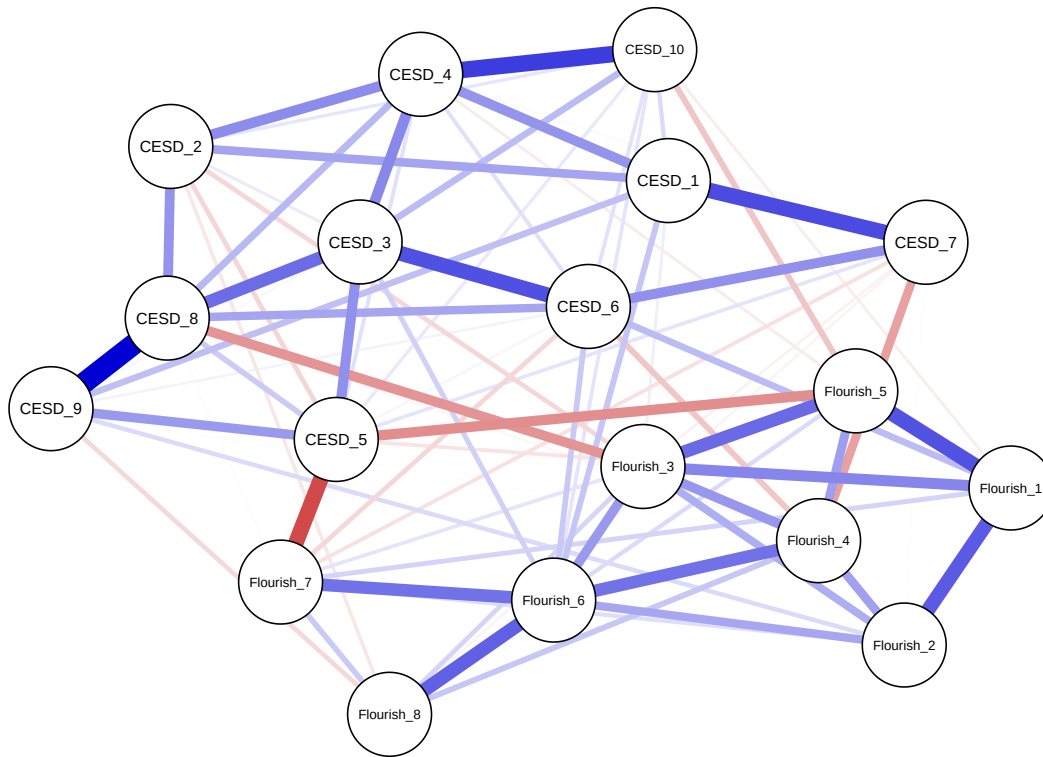
Based on the correlation analyses, both CESD_5 and CESD_8 show strong positive correlations with the total CES-D score (CESD_PreTotal), while their reverse-coded versions (CESD_5_R and CESD_8_R) show strong negative correlations.

This pattern indicates that the original CESD_5 and CESD_8 variables were already reverse-coded in the raw dataset. Applying reverse coding again ($3 - x$) in this case results in inverting the intended direction, leading to an incorrect interpretation.

Therefore, reverse coding is not necessary, as the original variables are already in the correct (reverse-coded) format.

#flourishing and CESD (Depression)

```
network_data3 <- PPI_dataset[, c("CESD_1", "CESD_2", "CESD_3", "CESD_4", "CESD_5",  
  "CESD_6", "CESD_7", "CESD_8", "CESD_9", "CESD_10", "Flourish_1", "Flourish_2", "Flourish_3", "Flourish_4", "Flourish_5", "Flourish_6", "Flourish_7", "Flourish_8", "Flourish_9", "Flourish_10", "Flourish_11", "Flourish_12", "Flourish_13", "Flourish_14", "Flourish_15", "Flourish_16", "Flourish_17", "Flourish_18", "Flourish_19", "Flourish_20", "Flourish_21", "Flourish_22", "Flourish_23", "Flourish_24", "Flourish_25", "Flourish_26", "Flourish_27", "Flourish_28", "Flourish_29", "Flourish_30", "Flourish_31", "Flourish_32", "Flourish_33", "Flourish_34", "Flourish_35", "Flourish_36", "Flourish_37", "Flourish_38", "Flourish_39", "Flourish_40", "Flourish_41", "Flourish_42", "Flourish_43", "Flourish_44", "Flourish_45", "Flourish_46", "Flourish_47", "Flourish_48", "Flourish_49", "Flourish_50", "Flourish_51", "Flourish_52", "Flourish_53", "Flourish_54", "Flourish_55", "Flourish_56", "Flourish_57", "Flourish_58", "Flourish_59", "Flourish_60", "Flourish_61", "Flourish_62", "Flourish_63", "Flourish_64", "Flourish_65", "Flourish_66", "Flourish_67", "Flourish_68", "Flourish_69", "Flourish_70", "Flourish_71", "Flourish_72", "Flourish_73", "Flourish_74", "Flourish_75", "Flourish_76", "Flourish_77", "Flourish_78", "Flourish_79", "Flourish_80", "Flourish_81", "Flourish_82", "Flourish_83", "Flourish_84", "Flourish_85", "Flourish_86", "Flourish_87", "Flourish_88", "Flourish_89", "Flourish_90", "Flourish_91", "Flourish_92", "Flourish_93", "Flourish_94", "Flourish_95", "Flourish_96", "Flourish_97", "Flourish_98", "Flourish_99", "Flourish_100")]  
  
network_data3 <- na.omit(network_data3)  
network_data3 <- network_data3[, sapply(network_data3, function(x) sd(x, na.rm = TRUE) != 0)]  
  
library(bootnet)  
library(qgraph)  
library(dplyr)  
  
network_model3 <- estimateNetwork(network_data3,  
  default = "EBICglasso",  
  corMethod = "cor_auto")  
  
## Estimating Network. Using package::function:  
## - qgraph::EBICglasso for EBIC model selection  
## - using glasso::glasso  
## - qgraph::cor_auto for correlation computation  
## - using lavaan::lavCor  
  
## Variables detected as ordinal: CESD_1; CESD_2; CESD_3; CESD_4; CESD_5; CESD_6; CESD_7; CESD_8; CESD_9; CESD_10; CESD_11; CESD_12; CESD_13; CESD_14; CESD_15; CESD_16; CESD_17; CESD_18; CESD_19; CESD_20; CESD_21; CESD_22; CESD_23; CESD_24; CESD_25; CESD_26; CESD_27; CESD_28; CESD_29; CESD_30; CESD_31; CESD_32; CESD_33; CESD_34; CESD_35; CESD_36; CESD_37; CESD_38; CESD_39; CESD_40; CESD_41; CESD_42; CESD_43; CESD_44; CESD_45; CESD_46; CESD_47; CESD_48; CESD_49; CESD_50; CESD_51; CESD_52; CESD_53; CESD_54; CESD_55; CESD_56; CESD_57; CESD_58; CESD_59; CESD_60; CESD_61; CESD_62; CESD_63; CESD_64; CESD_65; CESD_66; CESD_67; CESD_68; CESD_69; CESD_70; CESD_71; CESD_72; CESD_73; CESD_74; CESD_75; CESD_76; CESD_77; CESD_78; CESD_79; CESD_80; CESD_81; CESD_82; CESD_83; CESD_84; CESD_85; CESD_86; CESD_87; CESD_88; CESD_89; CESD_90; CESD_91; CESD_92; CESD_93; CESD_94; CESD_95; CESD_96; CESD_97; CESD_98; CESD_99; CESD_100; Flourish_1; Flourish_2; Flourish_3; Flourish_4; Flourish_5; Flourish_6; Flourish_7; Flourish_8; Flourish_9; Flourish_10; Flourish_11; Flourish_12; Flourish_13; Flourish_14; Flourish_15; Flourish_16; Flourish_17; Flourish_18; Flourish_19; Flourish_20; Flourish_21; Flourish_22; Flourish_23; Flourish_24; Flourish_25; Flourish_26; Flourish_27; Flourish_28; Flourish_29; Flourish_30; Flourish_31; Flourish_32; Flourish_33; Flourish_34; Flourish_35; Flourish_36; Flourish_37; Flourish_38; Flourish_39; Flourish_40; Flourish_41; Flourish_42; Flourish_43; Flourish_44; Flourish_45; Flourish_46; Flourish_47; Flourish_48; Flourish_49; Flourish_50; Flourish_51; Flourish_52; Flourish_53; Flourish_54; Flourish_55; Flourish_56; Flourish_57; Flourish_58; Flourish_59; Flourish_60; Flourish_61; Flourish_62; Flourish_63; Flourish_64; Flourish_65; Flourish_66; Flourish_67; Flourish_68; Flourish_69; Flourish_70; Flourish_71; Flourish_72; Flourish_73; Flourish_74; Flourish_75; Flourish_76; Flourish_77; Flourish_78; Flourish_79; Flourish_80; Flourish_81; Flourish_82; Flourish_83; Flourish_84; Flourish_85; Flourish_86; Flourish_87; Flourish_88; Flourish_89; Flourish_90; Flourish_91; Flourish_92; Flourish_93; Flourish_94; Flourish_95; Flourish_96; Flourish_97; Flourish_98; Flourish_99; Flourish_100)  
  
plot(network_model3)
```



```
#flourishing and Autonomy
```

```
network_data4 <- PPI_dataset[, c("PsyNeed_1", "PsyNeed_7", "PsyNeed_13", "PsyNeed_19", "Flourish_1", "Flourish_2", "Flourish_3", "Flourish_4", "Flourish_5", "Flourish_6", "Flourish_7", "Flourish_8")]
```

```
network_data4 <- na.omit(network_data4)
```

```
network_data4 <- network_data4[, sapply(network_data4, function(x) sd(x, na.rm = TRUE) != 0)]
```

```
library(bootnet)
```

```
library(qgraph)
```

```
library(dplyr)
```

```
network_model4 <- estimateNetwork(network_data4,
                                   default = "EBICglasso",
                                   corMethod = "cor_auto")
```

```
## Estimating Network. Using package::function:
```

```
## - qgraph::EBICglasso for EBIC model selection
```

```
## - using glasso::glasso
```

```
## - qgraph::cor_auto for correlation computation
```

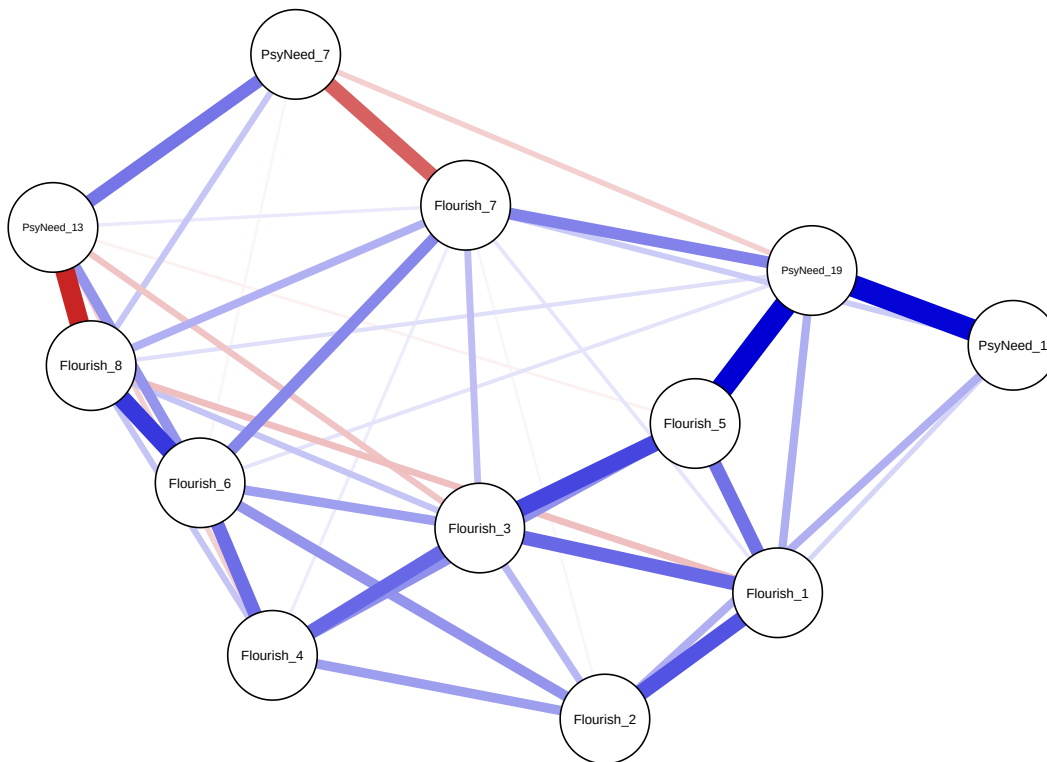
```
## - using lavaan::lavCor
```

```
## Variables detected as ordinal: PsyNeed_1; PsyNeed_7; PsyNeed_13; PsyNeed_19; Flourish_1; Flourish_2;
```

```
## Warning in EBICglassoCore(S = S, n = n, gamma = gamma, penalize.diagonal =
```

```
## penalize.diagonal, : A dense regularized network was selected ( $\lambda < 0.1 * \lambda_{\max}$ ). Recent work indicates a possible drop in specificity. Interpret the presence of the smallest edges with care. Setting threshold = TRUE will enforce higher specificity, at the cost of sensitivity.
```

```
plot(network_model4)
```



```
#flourishing and Relatedness
```

```
network_data5 <- PPI_dataset[, c("PsyNeed_3", "PsyNeed_9", "PsyNeed_15", "PsyNeed_21", "Flourish_1", "Flourish_2", "Flourish_3", "Flourish_4", "Flourish_5", "Flourish_6", "Flourish_7", "Flourish_8", "PsyNeed_13", "PsyNeed_19", "PsyNeed_1")]
```

```
network_data5 <- na.omit(network_data5)
```

```
network_data5 <- network_data5[, sapply(network_data5, function(x) sd(x, na.rm = TRUE) != 0)]
```

```
library(bootnet)
```

```
library(qgraph)
```

```
library(dplyr)
```

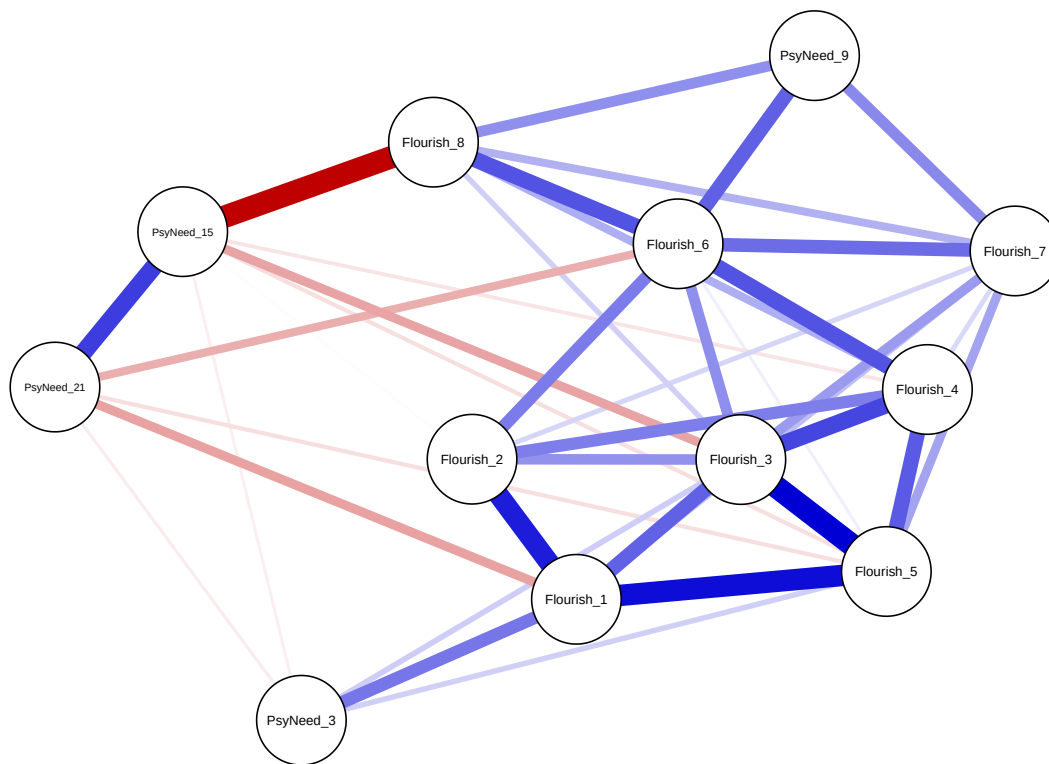
```
network_model5 <- estimateNetwork(network_data5,
                                   default = "EBICglasso",
                                   corMethod = "cor_auto")
```

```
## Estimating Network. Using package::function:
## - qgraph::EBICglasso for EBIC model selection
## - using glasso::glasso
```

```
## - qgraph::cor_auto for correlation computation
## - using lavaan::lavCor
```

```
## Variables detected as ordinal: PsyNeed_3; PsyNeed_9; PsyNeed_15; PsyNeed_21; Flourish_1; Flourish_2;
```

```
plot(network_model5)
```



```
#flourishing and Competence
```

```
network_data6 <- PPI_dataset[, c("PsyNeed_5", "PsyNeed_11", "PsyNeed_17", "PsyNeed_23", "Flourish_1",
```

```
network_data6 <- na.omit(network_data6)
```

```
network_data6 <- network_data6[, sapply(network_data6, function(x) sd(x, na.rm = TRUE) != 0)]
```

```
library(bootnet)
```

```
library(qgraph)
```

```
library(dplyr)
```

```
network_model6 <- estimateNetwork(network_data6,
                                   default = "EBICglasso",
                                   corMethod = "cor_auto")
```

```
## Estimating Network. Using package::function:
```

```
## - qgraph::EBICglasso for EBIC model selection
```

```
## - using glasso::glasso
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## - using lavaan::lavCor
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```
## Variables detected as ordinal: PsyNeed_5; PsyNeed_11; PsyNeed_17; PsyNeed_23; Flourish_1; Flourish_2
```

```
plot(network_model6)
```

